



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2012
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each Section (A, B, and C) is to be returned in a separate bundle.
- All necessary working should be shown in every question, except multiple choice.

Total Marks – 75

- Attempt questions 1 – 6.
- All questions are not of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, x > 0$

Section A (22 Marks)

START A NEW BOOKLET

Question 1. (7 marks)

Indicate which of the answers A, B, C, or D is the correct answer. Write the letter corresponding to the answer in your answer booklet. **Marks**

(a) Solve for y : $xy - d = m$:- **A:** $y = \frac{m-d}{x}$ **1**

B: $y = m + d - x$

C: $y = \frac{m+d}{x}$

D: $xy = m + d$

(b) Reduce the following expression to simplest form: **1**

$$\frac{12a^3c}{4ac} =$$

A: $8a^2$

B: $3a^2$

C: $3a^3$

D: $3a^3c$

(c) The derivative of $y = 6x^{-3}$ equals:- **1**

A: $-18x^{-4}$

B: $-18x^{-2}$

C: $-12x^{-3}$

D: $-3x^{-2}$

- (d) Find a primitive of $\frac{1}{2x+3}$:- **1**

A: $\frac{2}{(2x+3)^2}$

B: $\ln(2x+3)$

C: $\frac{1}{2}\ln(2x+3)$

D: $\ln(2x)+3$

- (e) If $(2, -3)$ is the midpoint of the interval joining $A(4, -2)$ and $B(x, y)$ then B has co-ordinates:- **1**

A: $(-4, 0)$

B: $(0, 0)$

C: $(0, -4)$

D: $(0, -8)$

- (f) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2x \, dx =$ **1**

A: $\frac{1}{2}$

B: 0

C: 1

D: $-\frac{1}{2}$

(g) The derivative of $\ln \sqrt{1-x^2}$ is:-

1

A: $\frac{1}{2\sqrt{1-x^2}}$

B: $\frac{-x}{1-x^2}$

C: $\frac{-x}{\sqrt{1-x^2}}$

D: $\frac{x}{\sqrt{x^2-1}}$

Question 2 (15 marks)

- | | |
|--|--------------|
| | Marks |
| (a) Differentiate the following: | 4 |
|
 | |
| (i) $y = 2 \tan 3x$ | |
| (ii) $y = x \sin x$ | |
| (iii) $y = e^{\sin x}$ | |
| (iv) $y = \ln 4x$ | |
|
 | |
| (b) Find | 4 |
|
 | |
| (i) $\int (1 - e^x)^2 dx$ | |
| (ii) $\int_{\frac{1}{6}}^{\frac{1}{2}} \cos \pi x dx$ | |
|
 | |
| (c) Use the trapezoidal rule with four function values to find an approximation to | 3 |
|
 | |
| $\int_{-2}^1 2^{-x} dx$ | |
| (Answer correct to two decimal places.) | |
|
 | |
| (d) The area enclosed between $y = 2x - x^2$ and $y = x$ is rotated about the x -axis. | 4 |
|
 | |
| (i) Find the points of intersection of the curves. | |
| (ii) Find the volume of the solid generated. | |

Section B (26 Marks)
START A NEW BOOKLET

Question 3 (13 Marks)

Marks

- (a) Find the area of a segment of a circle of radius 4 cm and angle at the centre 1.5 radians, correct to two decimal places.

3

- (b) Simplify: $\frac{1}{\sec^2 x} + \frac{1}{\operatorname{cosec}^2 x}$

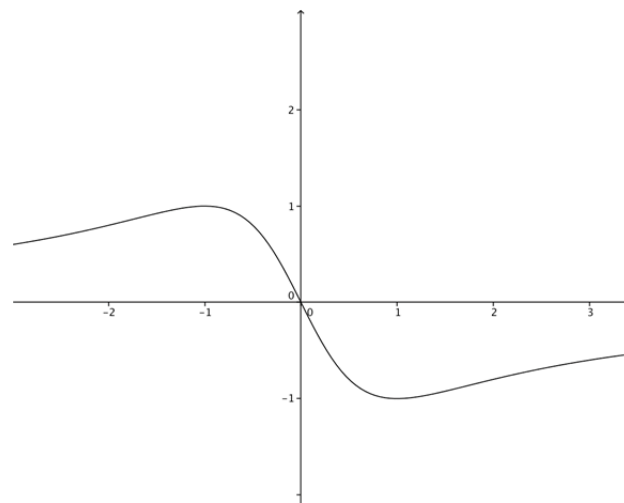
1

- (c) Sketch the graph of $y = 3\cos 2x$ in the domain $0 \leq x \leq 2\pi$.

2

- (d) The graph shows $f'(x)$, the derivative of $y = f(x)$.

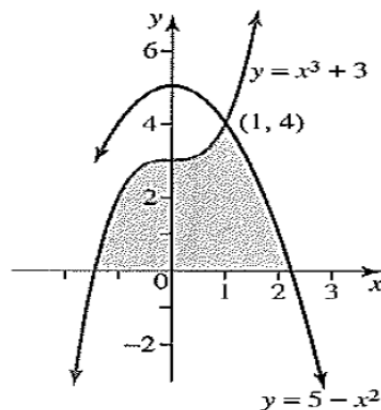
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- (i) Copy the graph to your answer booklet.
- (ii) Sketch on the graph a possible graph of $y = f(x)$ given that $f(x) > 0$ for all x . On your graph of $y = f(x)$ mark any points of inflexion.

- (e)

3



Find the area bounded by the curves $y = x^3 + 3$, $y = 5 - x^2$ and the x -axis.

Question 4 (13 Marks)

- (a) For a function $f(x)$ it is given that $f(0) = 4$ and $f(1) = 12$. **3**
Find, in simplest form, the value of:

$$\int_0^1 \frac{f'(x)}{f(x)} dx$$

- (b) Given the function $y = 2 + 3x - x^3$: **6**

- (i) Find the co-ordinates of the stationary points, and determine their nature.
- (ii) Find the co-ordinates of any points of inflexion.
- (iii) Sketch the curve in the domain $-3 \leq x \leq 3$.

- (c) Find the equation of the normal to the curve $y = e^{-x}$ at the point where $x = 1$. **2**

- (d) The Fundamental Theorem of the Integral Calculus is: **2**

$$\int_a^b f(x) dx = F(b) - F(a)$$

where $F(x)$ is a primitive of $f(x)$.

Use this theorem to prove that:

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Section C (27 Marks)

START A NEW BOOKLET

Question 5 (13 Marks)

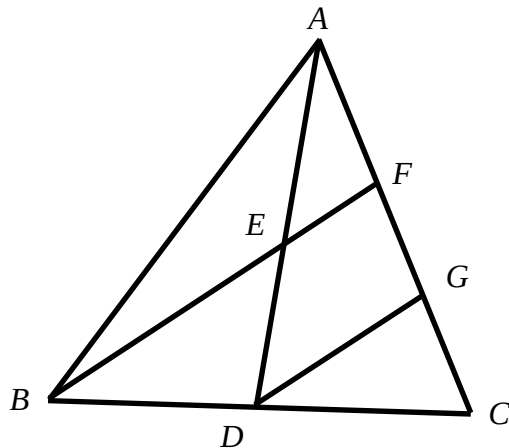
- (a) A sphere of volume $V = \frac{4}{3}\pi r^3$ and surface area $A = 4\pi r^2$ is expanding at a constant rate. Find the ratio of the rate of change of volume with respect to r to the rate of change of surface area with respect to r . 2

- (b) For the triangle with vertices $A(-3, 0)$, $B(2, 4)$, and $C(6, -1)$: 6

(i) Show that the triangle is isosceles.

(ii) Find the area of the triangle.

- (c) 5



In the diagram, D and E are the midpoints of BC and AD respectively, and $DG \parallel BF$.

- (i) Copy the diagram to your answer booklet.
- (ii) Prove that $AF = FG = GC$.

Question 6 (14 Marks)

- (a) Simplify $\ln(\ln \sqrt{e^4})$. 1
- (b) A particle is moving in a straight line with acceleration given by $a = 12t \text{ m/s}^2$. 4
Initially the particle is at the origin with velocity -8 m/s .
- (i) Find the equation for its position at time t .
- (ii) State when the particle is next at the origin.
- (c) (i) Find $\frac{d}{dx}\left(\frac{3}{x^3+1}\right)$. 4
- (ii) The region of the number plane under the curve $y = \frac{3x}{x^3+1}$, above the x -axis, and between $x = 0$ and $x = 1$ is rotated about the x -axis. With the aid of part (i), find then volume of the solid formed.
- (d) The volume of a closed cylindrical container is 800 m^3 . The container is to be made of aluminium sheeting costing \$78 per square metre. 5
- (i) Show that, given the radius of the base is r , the area of aluminium required is
- $$A = 2\pi r^2 + \frac{1600}{r}$$
- (ii) Find the minimum cost (to the nearest dollar) of the aluminium for making the container.

This is the end of the paper.

Q1.

a) C b) B c) A

d) C e) C f) C

g) B

$$b) ii) \int_{\frac{1}{6}}^{\frac{1}{2}} \cos \pi x \cdot dx$$

$$= \left[\frac{1}{\pi} \sin \pi x \right]_{\frac{1}{6}}^{\frac{1}{2}}$$

$$= \frac{1}{\pi} \left(\sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right)$$

$$= \frac{1}{\pi} \left(1 - \frac{1}{2} \right)$$

$$= \frac{1}{2\pi} //$$

Q2

$$a) i) y = 2 \tan 3x$$

$$y' = 6 \sec^2 3x$$

$$ii) y = x \sin x$$

$$y' = \sin x (1 + x \cos x)$$

$$iii) y = e^{\sin x}$$

$$y' = \cos x e^{\sin x}$$

$$iv) y = \ln 4x$$

$$y' = \frac{4}{4x} = \frac{1}{x}$$

$$c) h = \frac{b-a}{n} \quad n=3$$

$$= \frac{1 - (-2)}{3} = 1$$

$$\int_{-2}^1 2^{-x} dx = \frac{1}{2} (y_0 + y_3 + 2(y_1 + y_2))$$

x	y ₀	y ₁	y ₂	y ₃
-2	4	2	1	$\frac{1}{2}$

$$= \frac{1}{2} \left(4 + \frac{1}{2} + 2(2 + 1) \right)$$

$$= \frac{1}{2} \left(10 \frac{1}{2} \right)$$

$$\div 5.25$$

$$b) i) \int (1 - e^x)^2 \cdot dx$$

$$= \int (1 - 2e^x + e^{2x}) \cdot dx$$

$$= x - 2e^x + \frac{e^{2x}}{2} + C$$

$$d) V = \pi \int_0^1 (2x - x^2)^2 dx - \pi \int_0^1 x^2 \cdot dx$$

$$= \pi \int_0^1 (4x^2 - 4x^3 + x^4) \cdot dx - \pi \int_0^1 x^2 \cdot dx$$

$$= \pi \left[\frac{4}{3} x^3 - \frac{4}{4} x^4 + \frac{x^5}{5} \right] - \pi \left[\frac{x^3}{3} \right]$$

$$= \pi \left[\frac{4}{3} - 1 + \frac{1}{5} \right] - \pi \left[\frac{1}{3} \right] = 0$$

$$= \frac{\pi}{5} \text{ units}^3$$

D) Points of intersection

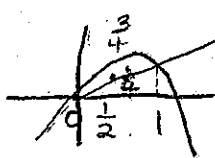
$$2x - x^2 = x$$

$$0 = x^2 - x$$

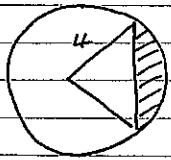
$$0 = x^2 - x$$

$$0 = x(x - 1)$$

$$x = 0 \quad x = 1$$



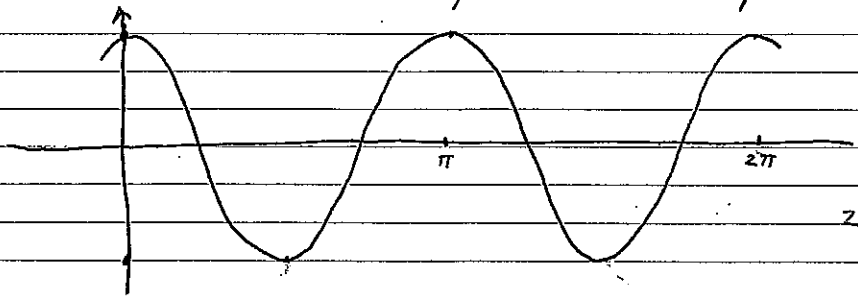
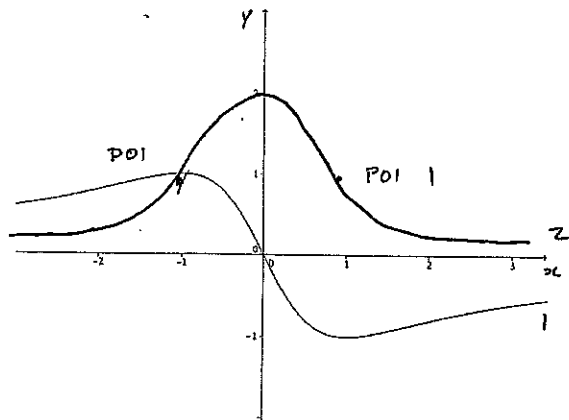
3(a)



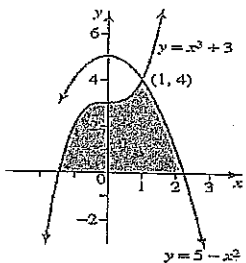
$$\begin{aligned}\text{Area of Minor Segment} &= \frac{1}{2}(r^2(\theta - \sin\theta)) \\ &= 8(1.5 - \sin 1.5) \\ &= 4.023\end{aligned}$$

(Major Segment = 46.25)

$$(b) \quad \frac{1}{\sec^2 x} + \frac{1}{\csc^2 x} = \cos^2 x + \sin^2 x = 1$$

(c) $y = 3 \cos 2x$ has amplitude 3 and period π .(d) (i)
(ii)

Question 3 (e)



$y = x^3 + 3$ cuts x-axis at $x = \sqrt[3]{3}$
 $y = 5 - x^2$ cuts x-axis at $x = \sqrt{5}$

$$\begin{aligned}\text{Area} &= \int_{-\sqrt{3}}^1 x^3 + 3 \cdot dx + \int_1^{\sqrt{5}} 5 - x^2 \cdot dx \\ &= \left[\frac{1}{4}x^4 + 3x \right]_{-\sqrt{3}}^1 + \left[5x - \frac{x^3}{3} \right]_1^{\sqrt{5}} \\ &= 6.50 + 2.79 = 9.29\end{aligned}$$

4 (a)

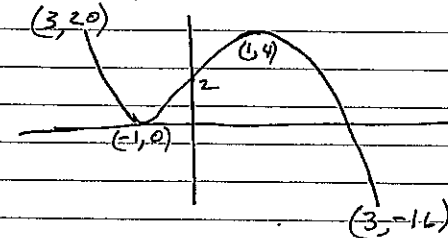
$$\begin{aligned}\int_0^1 \frac{f'(x)}{f(x)} \cdot dx &= [\ln f(x)]_0^1 \\ &= \ln f(1) - \ln f(0) \\ &= \ln 12 - \ln 4 \\ &= \ln 3\end{aligned}$$

$$(b) \quad \begin{aligned}y &= 2 + 3x - x^2 \\ \frac{dy}{dx} &= 3 - 2x^2 \\ \frac{d^2y}{dx^2} &= -4x\end{aligned}$$

Stat PG. $\frac{dy}{dx} = 0$

$$\therefore 3 - 2x^2 = 0$$

$$x = \pm 1$$

at $x = 1$; $\frac{d^2y}{dx^2}$ is -ve $\therefore (1, 4)$ is Max. Turn Ptat $x = -1$; $\frac{d^2y}{dx^2}$ is +ve $\therefore (-1, 0)$ is Min. Turn PtP.O.I at $\frac{d^2y}{dx^2} = 0 \therefore$ at $x = 0$ Before $x = 0$ $\frac{d^2y}{dx^2}$ is +ve and after $x = 0$ $\frac{d^2y}{dx^2}$ is -ve $\therefore$ Concavity changes $(0, 2)$ is P.O.I

$$(c) \quad \begin{aligned}y &= e^{-x} \\ \frac{dy}{dx} &= -e^{-x}\end{aligned}$$

at $x = 1$, $y = \frac{1}{e}$ and gradient of normal is $= \frac{1}{-e^{-1}} = e$ $\therefore$ Eqn is $y - \frac{1}{e} = e(x - 1)$

$$ey - 1 = e^2(x - 1)$$

$$ey - e^2x - 1 + e^2 = 0$$

4(d) Prove $\int_0^a f(x) \cdot dx = \int_0^a f(a-x) \cdot dx$

$$\text{LHS} = F(a) - F(0)$$

$$\text{RHS} = [-F(a-x)]_0^a$$

$$= -F(0) + F(a)$$

$$= F(a) - F(0)$$

$$= \text{LHS}$$

Q.E.D.

Question 5

a) $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dr} = 4\pi r^2$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dV}{dr} : \frac{dA}{dr}$$

$$= 4\pi r^2 : 8\pi r$$

$$= r : 2$$

b) i) $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$AB = \sqrt{(2 - (-3))^2 + (4 - 0)^2}$$
$$= \sqrt{41}$$

$$AC = \sqrt{(6 - (-3))^2 + (-1 - 0)^2}$$
$$= \sqrt{82}$$

$$BC = \sqrt{(2 - 6)^2 + (4 - (-1))^2}$$
$$= \sqrt{41}$$

since $AB = BC$

$\therefore \triangle ABC$ is isosceles

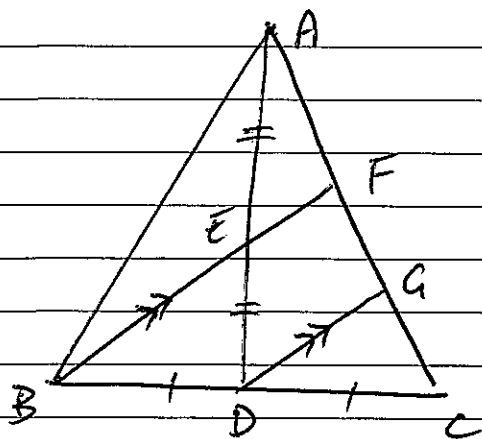
ii) since pythagoras' theorem holds
ie $AB^2 + BC^2 = AC^2$

$\triangle ABC$ is right angled

$$\therefore A = \frac{1}{2} \times \sqrt{41} \times \sqrt{41}$$

$$= \frac{41}{2} \text{ units}^2$$

c) i)



ii) $\frac{FG}{GC} = \frac{BD}{DC}$ (ratio of intercepts, $DG \parallel BF$)

$BD = DC$ (D is midpoint of BC)

$\therefore \frac{FG}{GC} = 1$

$FG = GC$

$\frac{AF}{FG} = \frac{AE}{ED}$ (ratio of intercepts, $DG \parallel BF$)

$AE = ED$ (E is midpoint of AD)

$\therefore \frac{AF}{FG} = 1$

$AF = FG$

And so $AF = FG = GC$

Question 6

a) $\ln(\ln \sqrt{e^4})$

$= \ln(\ln(e^4)^{\frac{1}{2}})$

$= \ln(\ln e^2)$

$= \ln 2$

b) i) $a = 12t$

$v = 6t^2 + C$

when $t = 0, v = -8$

$-8 = C$

$$v = 6t^2 - 8$$

$$x = 2t^3 - 8t + C$$

$$\text{when } t=0, x=0$$

$$C=0$$

$$\therefore x = 2t^3 - 8t \quad \text{m.}$$

$$\text{ii) when } x=0$$

$$0 = 2t^3 - 8t$$

$$0 = 2t(t^2 - 4)$$

$$0 = 2t(t-2)(t+2)$$

$$t = -2, 0, 2$$

particle is next at the origin after 2 seconds.

$$\text{c) i) } \frac{d}{dx} \left(\frac{3}{x^3+1} \right)$$

$$= \frac{d}{dx} \left(3(x^3+1)^{-1} \right)$$

$$= -3(x^3+1)^{-2} \cdot 3x^2$$

$$= \frac{-9x^2}{(x^3+1)^2}$$

$$\text{ii) } V = \pi \int_a^b y^2 dx$$

$$V = \pi \int_0^1 \left(\frac{3x}{x^3+1} \right)^2 dx$$

$$= \pi \int_0^1 \frac{9x^2}{(x^3+1)^2} dx$$

$$= -\pi \int_0^1 \frac{-9x^2}{(x^3+1)^2} dx$$

$$= -\pi \left[\frac{3}{x^3+1} \right]_0^1$$

$$= -\pi \left[\frac{3}{(1)^3+1} - \frac{3}{(0)^3+1} \right]$$

$$= \frac{3\pi}{2} \text{ units}^3$$

$$\begin{aligned}
 \text{a) i) } v &= \pi r^2 h \\
 \pi r^2 h &= 800 \\
 h &= \frac{800}{\pi r^2} \quad \text{--- (1)}
 \end{aligned}$$

$$\begin{aligned}
 A &= 2\pi r^2 + 2\pi r h \quad \text{--- (2)} \\
 \text{sub (1) into (2)}
 \end{aligned}$$

$$A = 2\pi r^2 + 2\pi r \left(\frac{800}{\pi r^2} \right)$$

$$A = 2\pi r^2 + \frac{1600}{r}$$

$$\begin{aligned}
 \text{ii) } A &= 2\pi r^2 + 1600r^{-1} \\
 \frac{dA}{dr} &= 4\pi r - 1600r^{-2} \\
 \frac{d^2A}{dr^2} &= 4\pi + 3200r^{-3}
 \end{aligned}$$

For stat. points let $\frac{dA}{dr} = 0$

$$4\pi r - \frac{1600}{r^2} = 0$$

$$4\pi r^3 - 1600 = 0$$

$$4\pi r^3 = 1600$$

$$r^3 = \frac{400}{\pi}$$

$$r = \sqrt[3]{\frac{400}{\pi}}$$

$$\frac{d^2A}{dr^2} = 4\pi + \frac{3200}{r^3}$$

clearly $\frac{d^2A}{dr^2} > 0$ when $r > 0$

$\therefore$ Minimum A when $r = \sqrt[3]{\frac{400}{\pi}}$

$$\text{Minimum } A = 2\pi \left(\sqrt[3]{\frac{400}{\pi}} \right)^2 + \frac{1600}{\left(\sqrt[3]{\frac{400}{\pi}} \right)}$$

$$\begin{aligned}
 \text{Minimum Cost} &= \left(2\pi \left(\sqrt[3]{\frac{400}{\pi}} \right)^2 + \frac{1600}{\left(\sqrt[3]{\frac{400}{\pi}} \right)} \right) \times 78 \\
 &= \$37211 \quad (\text{to nearest dollar})
 \end{aligned}$$